

What we found

The analysis used unsupervised methods to identify unusual insurance claims without using the fraud label during model fitting. K-Means identified six relatively stable claim groups, while isolation forest created a ranking of unusual claims for manual review.

The six cluster solution was highly stable across five random seeds, with Adjusted Rand Index values of approximately 0.996 to 0.999 against the first run. The anomaly ranking was less stable: between 78 and 85 of the top 100 claims overlapped with the first run.

The overall fraud rate in the dataset was 3.69%. The isolation forest anomaly ranking achieved an AUC of approximately 0.668 on the full dataset, and about 9% of the top 100 most unusual claims were fraudulent. Therefore most claims in the review queue were not fraudulent.

How confident we are

There is evidence that the clustering result is stable because the cluster assignments changed very little across random seeds. The anomaly ranking also showed substantial overlap between runs, although individual claims can move in and out of the top 100.

The null test showed that clustering can still produce a non-zero silhouette score after feature relationships are destroyed. This demonstrates that a clustering metric alone is not enough to establish that meaningful structure exists.

For anomaly detection, the null case produced an AUC close to 0.50, indicating that the detector did not retain meaningful fraud-ranking ability when the feature relationships were removed. The main anomaly result also survived the sensitivity check using a different scaling method.

What the result does not supports

The anomaly score does not identify fraud directly.

Many unusual claims are legitimate. In the top 100 claims, most were not fraudulent. The method can also miss fraudulent claims that look normal according to the available features.

The dataset does not contain a customer identifier, so repeated claims from the same customer cannot be ruled out. This limits the independence assessment.

Performance also varies by subgroup. The online channel was the weakest evaluated subgroup.

Recommendation

Use the anomaly ranking as a manual review queue rather than as an automatic fraud decision. Claims with the highest anomaly scores can be reviewed first, while investigators make the final decision using additional information.

Before deployment

Before deployment, the system should be tested on newly confirmed claims over a future time period. The review team should monitor Precision@100 and check whether performance remains stable across claims channels, regions and policy types.

The anomaly ranking should be re-evaluated regularly because changes in claim behaviour can change what is considered unusual.